SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –** SIH26051
- **Problem Statement Title –** Software Based Model Development for Design of Area Specific Shelter for Thermal Comfort Maintenance
- **Theme –** Miscellaneous
- **PS Category –** Software
- **Team ID –** [team ID]
- **Team Name (Registered on portal) –** ASPIRE

ASPIRE

A climate-specific digital model for designing passive shelters in Ladakh

SMART INDIA HACKATHON 2026

What the problem statement is asking us to build

- A software model that predicts shelter temperature for a chosen outdoor climate.
- A user can change shelter size, shape, orientation, openings, material and thickness.
- Real weather and solar-radiation data are used instead of fixed assumptions.
- Different designs are compared under the same ambient conditions.
- The final output recommends the combination that maintains comfort with less active heating.

Why Ladakh?

- Large day/night temperature variation.
- Strong solar resource makes passive solar gain useful.
- Heat loss through walls, roofs and openings is significant.
- Thermal mass + insulation can store daytime heat and slow night-time losses.

Core design principle

SOLAR GAIN

Capture useful winter sunlight

THERMAL MASS

Store heat and release it later

INSULATION

Reduce conductive heat loss

ORIENTATION

Use site-specific solar exposure

Three required outputs

01

Indoor temperature

T_inside(t) for the selected geometry and materials

02

Solar energy

Heat gained from incident solar radiation

03

Heat flow

Heat entering/leaving through the envelope over time

ASPIRE

Technical Approach

Fast computational screening + ANSYS validation + interactive visualization

Thermal simulation pipeline

1. CLIMATE INPUT
NASA POWER hourly temperature, solar radiation, wind

2. USER DESIGN
Dimensions, orientation, openings, materials

3. PYTHON MODEL
U-value + thermal mass + solar gain + RK4

4. ANSYS
Detailed thermal simulation, mesh + boundary conditions

5. COMPARISON
Materials, thickness, shape and orientation

6 RECOMMENDATION
Best design for comfort and low heat demand

Technology stack

Engineering	ANSYS Workbench • transient thermal • parametric studies
Scientific Python	NumPy • Pandas • RK4 solver • U/R-value • thermal mass • heat balance
Backend	Flask • Flask-CORS • REST API • JSON simulation endpoints
Frontend	HTML/CSS/JavaScript • Chart.js • Three.js • OrbitControls
Data layer	NASA POWER hourly data • material-property CSV • optional SQLite/PostgreSQL
Sensors / IoT	ESP32/Arduino • temperature • humidity • solar irradiance • wind
Deployment	Git/GitHub • Docker • cached/offline datasets for remote sites

ASPIRE

Feasibility And Viability

SMART INDIA HACKATHON 2026

Why the solution is practical — and how we will control technical and deployment risk

Technical feasibility

What already works

- 1,488 hourly weather records are already available in the supplied dataset.
- Python backend already performs thermal calculations and exposes API routes.
- Material library supports multiple envelope options for comparison.
- Chart.js and Three.js already provide result visualization and 3D interaction.

Next validation step

- Create an ANSYS reference geometry.
- Match the same ambient conditions and material properties.
- Compare ANSYS temperature/heat-flux results with the fast Python model.

Operational feasibility

Why it is viable

- Most inputs are numeric and can be entered through a simple form.
- The fast model can screen many designs before expensive simulation.
- The system can work with historical data first and live sensors later.
- The architecture can be reused for other climatic regions.

Practical deployment

- Local cached weather data keeps the application usable with weak internet.
- Sensor correction can reduce the gap between satellite data and site reality.
- Saved simulation cases make material/design comparison repeatable.

Risk & mitigation

Main risks

- Real shelters have infiltration, thermal bridges and workmanship effects.
- Material properties can vary with moisture and temperature.
- ANSYS accuracy depends on geometry, mesh and boundary conditions.
- A single weather point cannot represent every Ladakh micro-climate.

How we mitigate them

- Sensitivity analysis across material k, thickness and opening area.
- Field calibration when sensor data becomes available.
- Human-in-the-loop review of shortlisted designs.
- Report uncertainty instead of presenting one number as absolute truth.

FEASIBILITY SCORECARD									
Software prototype	READY	Weather data	READY	Material comparison	READY	ANSYS validation	NEXT	Live sensors	FUTURE

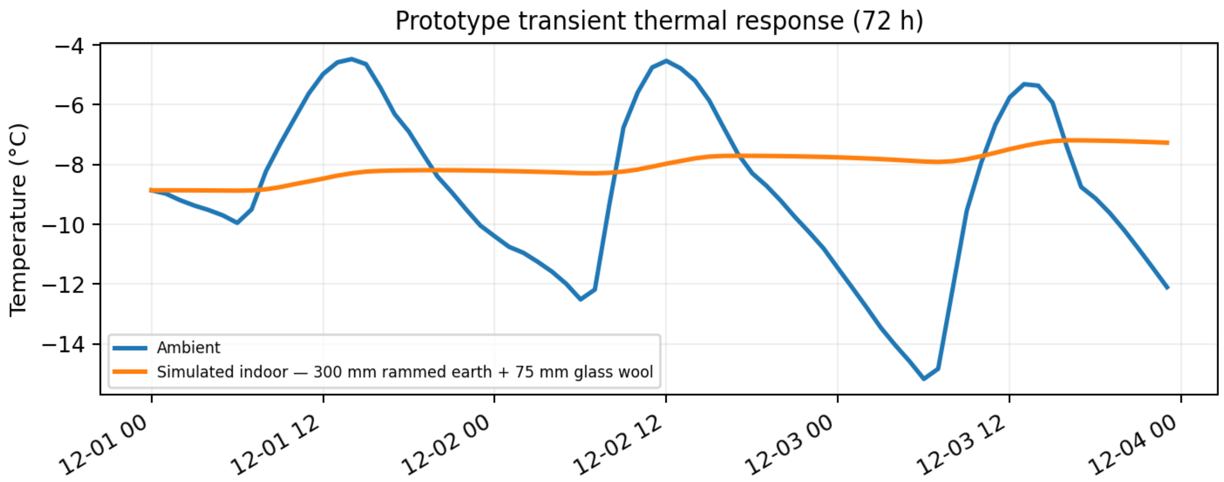
ASPIRE

Impact And Benefits

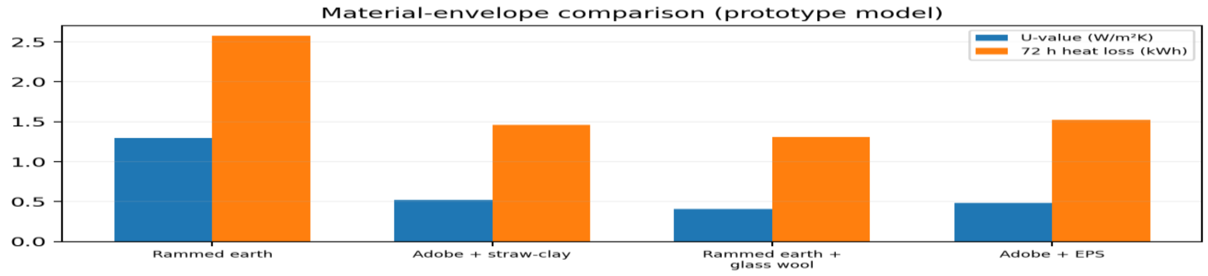
SMART INDIA HACKATHON 2026

Turn simulation results into a practical shelter-design decision

Prototype thermal response



Illustrative run using the supplied Ladakh weather data. The fast model is intended for screening; ANSYS and field measurements provide validation.



Key benefits

Energy Less dependence on active heating through better passive design.

Design Compare material, thickness, glazing and orientation before construction.

Cost Reduce trial-and-error physical prototyping.

Field use Accept local sensor readings and calibrate the model.

Scale Adapt the same framework to other climatic regions.

SUCCESS = more time in comfort band + less active energy demand

ASPIRE

Research & Reference

Climate data, passive-solar research and engineering-simulation references

SMART INDIA HACKATHON 2026

NASA POWER

Hourly solar and meteorological data for the selected location.

https://power.larc.nasa.gov/docs/services/api/temporal/hourly/

ANSYS Workbench

Thermal simulation and engineering workflow for high-fidelity validation.

https://www.ansys.com/en-in/products/ansys-workbench

Leh Climatological Data (2011–2024)

Regional temperature and weather records.

chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://mausam.imd.gov.in/leh/mcdata/City.pdf

FHNW — PSH Ladakh

Research project on passive-solar heated houses in Ladakh.

https://www.fhnw.ch/en/engineering-environment/research-services/research/projects/passive-solar-heated-houses-psh-in-ladakh-india

Khan (2013)

Vernacular Architecture & Climatic Control in Ladakh

https://www.researchgate.net/publication/279186605_Vernacular_Architecture_and_Climatic_Control_in_the_extreme_conditions_of_Ladakh_ASPIRE-2013

Mehta, Gajjar & Bhavsar (2024)

Thermal Comfort in Vernacular Ladakhi Dwellings

https://arch.su.ac.th/images/publications/ISVS12/20-Dinal.pdf

What we will demonstrate

- User changes shelter geometry and materials.
- Hourly climate data enters the model.
- Python rapidly screens candidate designs.
- ANSYS validates the shortlisted geometry.
- Graphs show indoor temperature, solar gain and heat flow.
- The system ranks designs against thermal-comfort targets.
- Future sensor integration makes the model site-specific.

THERMAL LADAKH

Passive-first • data-driven • region-specific

External references are linked above.